

MOI HIGH-KABARAK 2025 PREDICTION



Kenya Certificate of Secondary Education

KCSE PREDICTION SERIES 1



231/3

Biology (Practical)

Paper

3

Time: 1½ Hours

SCHOOL.....

SIGN.....

Kenya Certificate of Secondary Education.

CONFIDENTIAL.

Each candidate should be provided with the following:

- 10ml of Solution Q- A mixture of glucose, egg albumin and ammonia solution
(20gms of glucose + 10gms of egg albumin in a liter of 1M ammonia solution)
- Distilled water in a 100ml beaker.
- Source of heat
- 2 test tubes.
- 10% sodium hydroxide solution.
- A glass rod.
- Visking tubing-8cm long.
- 1% copper sulphate solution.
- 2 threads/strings-about 5cm long.
- Benedict's solution.
- 2 labels
- A syringe enough to measure 10mls.
- 3 droppers

NB-All photographs should be coloured

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NAME.....

SCHOOL..... SIGN.....

INDEX NO..... ADM NO.....

Kenya Certificate of Secondary Education.

INSTRUCTIONS TO CANDIDATES

Answer all questions.

- You are supposed to spend the first 15 minutes of the 1 ¾ hours allowed for this paper reading the whole paper carefully before commencing your work.
- Answers must be written in the spaces provided on the question paper
- Answers should be written in English.
- Candidates may be penalized for recording irrelevant information and incorrect spelling especially of technical terms.

FOR EXAMINER'S USE ONLY

Question	Maximum Score	Candidate's Score
1	14	
2	16	
3	10	

Total score	40	
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1. You are provided with:

- Solution Q
- Distilled water in 100 ml beaker
- Visking tubing and two threads
- Glass rod
- 2 test tubes and two labels
- Benedict's solution 10% sodium hydroxide solution and 1% copper sulphate solution.

PROCEDURE:

- Tie the visking tubing tightly at one end using the thread and put about 5ml of solution Q. Tie tightly the other end ensuring that there is no leakage.
- Place the visking tubing in the distilled water in 100ml beaker and support it by tying the thread on a glass rod over the beaker.
- Allow the set up to stand for 30mins. Observe any changes in the visking tubing.
- Remove the visking tubing from the distilled water.
- Using a syringe draw about 5ml of the contents in the beaker labeled distilled water. Put it in the test tube and label it G. Repeat the same procedure to obtain 5ml of solution in the visking tubing, put in another test tube and label it H.

a) Using the reagents provided, carry out food tests to determine the components in substance **G** and **H** and fill the table below.

Substance	Food Test	Procedure	Observation	Conclusion

G				
H				

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8marks

b) Substance Q is a urine sample obtained from a patient in a hospital laboratory. Name two conditions that the patient was suffering from. **(2marks)**

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c) (i) What physiological activities that took place in the experiment you undertook above. **(2mark)**

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Account for your answer in C (i) above. **(2marks)**

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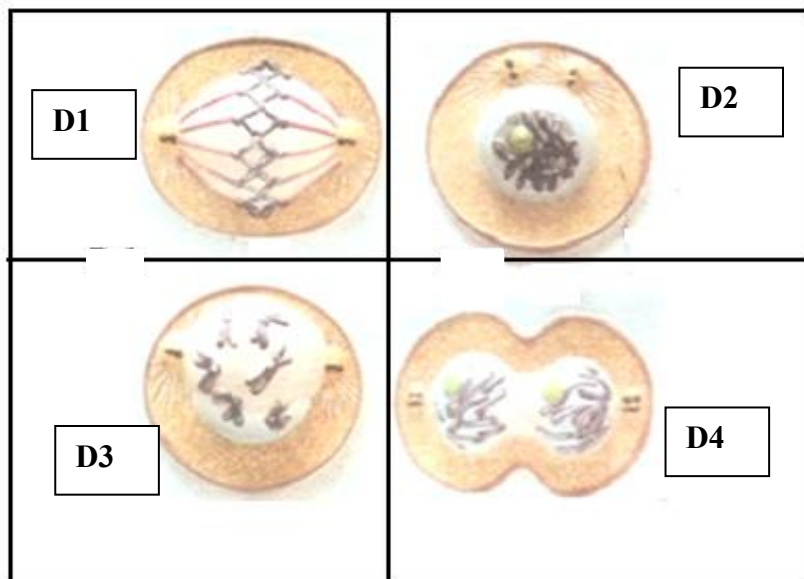
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2. Study the photomicrographs below and use them to answer questions that follow.



a) Identify the type of cell division above.

(1mark)

.....

b) Give a reason for your answer (i) above.

(1mark)

.....

.....

c) With a reason identify each stage.

(4marks)

Diagram	Stage	Reason
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D1		
D2		
D3		
D4		

(d) The photographs below show two fruits and their half-sections.



(i) Through which plane has the sections been cut?

(1mark)

.....

(ii) State two differences between specimens S and T. (2marks)

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.....
.....

(iii) With reasons, in each case state the type of fruit and method of dispersals for specimen S and T. (2marks)

Specimen S

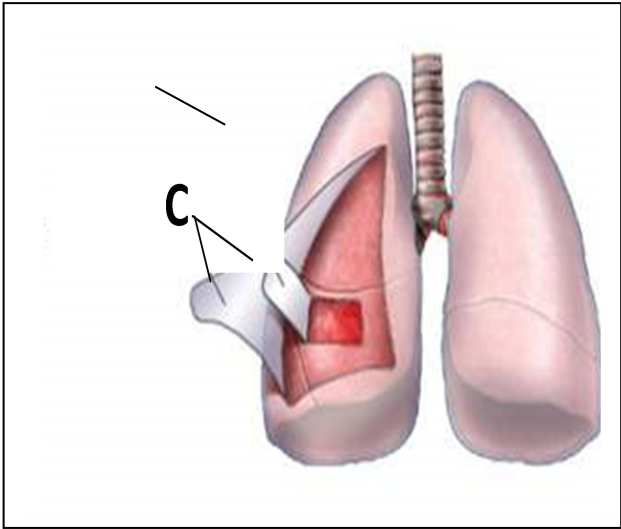
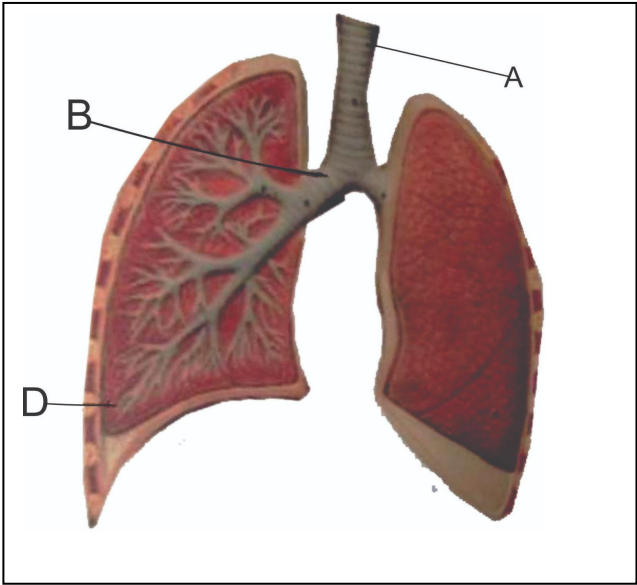
Type of fruit:
Reason:.....
Method of dispersal:.....
Reason:.....

Specimen T (2mks)

Type of fruit:.....
Reason:.....
Method of dispersal:.....
Reason.....

(iv) Draw and label a diagram of cut specimens S. (4marks)

3, Study the diagrams below and answer the questions that follow.



a) (i) Name the parts labeled A, B, C and D. (4marks)

A.....
B.....
C.....
D.....

(ii) State the adaptation of the part labeled A to its function. (2marks)

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(iii) State **one** function of the part labeled C. (1mark)

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b) State **three** adaptations of the part labeled D. (3marks)

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